

Ionospheric Reflections and Over-The-Horizon Naval Radar

ELEC-H415 - Chapter 5

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2026

Context: Over-The-Horizon (OTH) Threat Detection

The Line-of-Sight Microwave Horizon

Standard naval microwave radars ($\sim 3 \text{ GHz} - 10 \text{ GHz}$) are limited by Earth curvature:

$$d_{\text{LOS}} \approx \sqrt{2k_e R_e} \left(\sqrt{h_{\text{mast}}} + \sqrt{h_{\text{target}}} \right) \approx 30 - 50 \text{ km}$$

A warship cannot detect threats beyond this optical horizon.

The Skywave Reflection Radar

Over-The-Horizon (OTH) Radar exploits total reflection on the **ionosphere** in a lower frequency band, achieving detection ranges up to 4000 km.

Theoretical Goals of this Lesson:

- Plasma frequency & electron density profile $n_e(h)$.
- Total Internal Reflection condition via Snell's Law.
- Prove the Secant Law ($f_{\text{max}} = f_p / \cos \theta_i$).
- Analyze the Skip Zone & Tropospheric Limits.

Ionospheric Plasma & Electron Density Profile $n_e(h)$

Plasma Relative Permittivity

The ionosphere acts as a plasma layer with relative permittivity:

$$\varepsilon_r(\omega) = 1 - \frac{\omega_p^2}{\omega^2} < 1 \quad \text{with} \quad \omega_p = \sqrt{\frac{n_e q_e^2}{m_e \varepsilon_0}}$$

- $n_e(h)$: Free electron volume density [m^{-3}].
- Its variation is caused by solar activity.

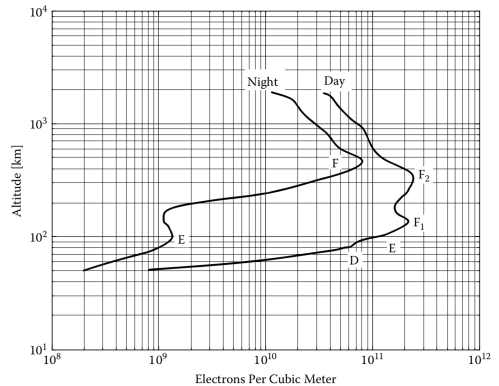


Figure: Electron density n_e in the ionosphere (Source : V.L. Granatstein, *Physical principles of wireless communications*, Auerbach, 2008)

Fundamental Plasma Frequency & Plasma Refractive Index

Substituting physical constants into $f_p = \frac{\omega_p}{2\pi}$:

- Elementary charge: $q_e \approx 1.602\,176\,63 \times 10^{-19} \text{ C}$
- Electron mass: $m_e \approx 9.109\,383\,7 \times 10^{-31} \text{ kg}$
- Vacuum permittivity: $\epsilon_0 \approx 8.854\,187\,8 \times 10^{-12} \text{ F m}^{-1}$

$$\begin{aligned} f_p &= \frac{1}{2\pi} \sqrt{\frac{q_e^2}{m_e \epsilon_0}} \sqrt{n_e} = \frac{1}{2\pi} \sqrt{\frac{(1.60218 \times 10^{-19})^2}{9.10938 \times 10^{-31} \times 8.85419 \times 10^{-12}}} \sqrt{n_e} \\ &= \frac{1}{2\pi} \sqrt{3.1826 \times 10^3} \sqrt{n_e} = \frac{56.415}{2\pi} \sqrt{n_e} \approx 8.9788 \sqrt{n_e} \\ &\implies f_p \approx 8.98 \sqrt{n_e} \text{ [Hz]} \quad (\text{with } n_e \text{ in } [\text{m}^{-3}]) \end{aligned}$$

Refractive Index of the Plasma

Assuming non-magnetic medium ($\mu = \mu_0$):

$$n = \frac{c}{v} = \frac{\sqrt{\epsilon \mu_0}}{\sqrt{\epsilon_0 \mu_0}} = \sqrt{\epsilon_r} = \sqrt{1 - \frac{\omega_p^2}{\omega^2}} = \sqrt{1 - \frac{f_p^2}{f^2}} < 1$$

Because $n < 1$, the ionosphere is an **optically less dense medium** than the neutral troposphere ($n_{\text{air}} \approx 1$).

Total Internal Reflection at the Ionospheric Boundary

Consider an EM wave incident on the ionospheric layer:

- **Below the ionosphere:** $\epsilon_{r1} \approx 1 \implies n_1 \approx 1$
- **In the ionosphere:** $\epsilon_{r2} = 1 - \frac{\omega_p^2}{\omega^2} < 1 \implies n_2 < n_1$

We can use **Snell's Law** at the interface:

$$n_2 \sin \theta_2 = n_1 \sin \theta_1 \implies \sin \theta_2 = \frac{n_1}{n_2} \sin \theta_i = \frac{\sin \theta_i}{\sqrt{\epsilon_r}}$$

Since $n_2 < n_1$, $\sin \theta_2$ can become ****greater than 1****!

Perpendicular Reflection Coefficient:

$$\Gamma_{\perp}(\theta_i) = \frac{\cos \theta_i - \sqrt{\epsilon_r - \sin^2 \theta_i}}{\cos \theta_i + \sqrt{\epsilon_r - \sin^2 \theta_i}}$$

Critical Angle θ_c

Total reflection happens when the transmitted wave angle reaches $\theta_2 = \pi/2$ ($\sin \theta_2 = 1$):

$$\sin \theta_c = \sqrt{\frac{\epsilon_2}{\epsilon_1}} = \sqrt{1 - \frac{\omega_p^2}{\omega^2}} = \sqrt{1 - \frac{f_p^2}{f^2}}$$

Condition for Total Reflection

Total reflection happens whenever $\theta_i \geq \theta_c$:

$$\sin \theta_i \geq \sqrt{1 - \frac{f_p^2}{f^2}}$$

Derivation of the Maximum Usable Frequency

From the total reflection condition:

$$\sin^2 \theta_i \geq 1 - \frac{f_p^2}{f^2}$$

Rearranging:

$$\frac{f_p^2}{f^2} \geq 1 - \sin^2 \theta_i = \cos^2 \theta_i$$

Taking the square root (since all frequencies and cosines are non-negative):

$$\frac{f_p}{f} \geq \cos \theta_i \iff f \leq \frac{f_p}{\cos \theta_i} = f_p \sec \theta_i$$

The Secant Law for OTH Radar

For a given incidence angle θ_i , total reflection occurs for all frequencies up to:

$$f_{\max} = \frac{f_{p,\max}}{\cos \theta_i}$$

- **Vertical Incidence** ($\theta_i = 0^\circ \implies \cos \theta_i = 1$): $f_{\max} = f_p$.
- **Oblique Naval Incidence** ($\theta_i \rightarrow 80^\circ \implies \cos \theta_i \approx 0.17$): $f_{\max} \approx 5.8 \times f_p \approx 20 - 30$ MHz.

Spherical Earth Geometry for Grazing Single-Hop Detection

To maximize detection range, the warship radiates at grazing elevation ($E \approx 0^\circ$, tangent to Earth):

1. **Angle of Incidence θ_i at Height h_i :** From the right-angle triangle ($\triangle OTI$):

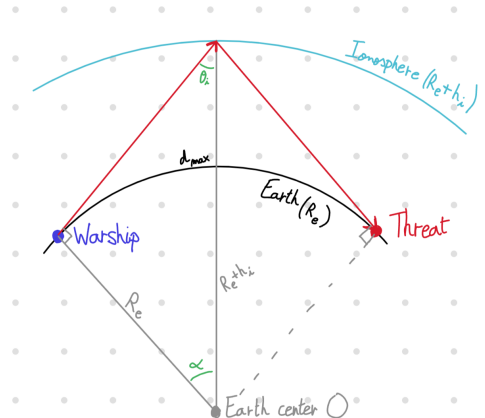
$$\sin \theta_i = \frac{R_e}{R_e + h_i} \quad (R_e = 6375 \text{ km})$$

2. **Angle at Earth Center α :**

$$\alpha = \frac{\pi}{2} - \theta_i$$

3. **Maximal Single-Hop Ground Arc $d_{\max}$:**

$$d_{\max} = 2R_e\alpha = 2R_e\left(\frac{\pi}{2} - \theta_i\right)$$



Applied Day vs. Night Detection Range Comparison

Daytime

High solar activity:

- Layer height: $h_i \approx 120 \text{ km}$
- Peak electron density: $n_{e,\max} \approx 2 \times 10^{11} \text{ m}^{-3}$

Step-by-Step Derivation:

- 1 $f_{p,\max} = 8.98\sqrt{2 \times 10^{11}} = 4.02 \text{ MHz}$
- 2 $\sin \theta_i = \frac{6375}{6495} = 0.9815 \implies \theta_i = 78.97^\circ$
- 3 $f_{\max} = \frac{4.02}{\cos(78.97^\circ)} = \frac{4.02}{0.1913} = 21.0 \text{ MHz}$
- 4 $d_{\max} = 2 \times 6375 \left(\frac{\pi}{2} - 1.3783 \right) = 2248 \text{ km}$

Nighttime

Low solar activity:

- Layer height: $h_i \approx 450 \text{ km}$
- Peak electron density: $n_{e,\max} \approx 8 \times 10^{10} \text{ m}^{-3}$

Step-by-Step Derivation:

- 1 $f_{p,\max} = 8.98\sqrt{8 \times 10^{10}} = 2.54 \text{ MHz}$
- 2 $\sin \theta_i = \frac{6375}{6825} = 0.9341 \implies \theta_i = 69.08^\circ$
- 3 $f_{\max} = \frac{2.54}{\cos(69.08^\circ)} = \frac{2.54}{0.3571} = 7.11 \text{ MHz}$
- 4 $d_{\max} = 2 \times 6375 \left(\frac{\pi}{2} - 1.2057 \right) = 4650 \text{ km}$

Time of Day Conclusion

Nighttime doubles the single-hop range (4650 km), but the radar frequency must be lowered to 7.1 MHz to prevent the wave from escaping into space.

Elevation Angle E to Incidence Angle θ_i Relationship

For non-grazing, with elevation angle $E > 0^\circ$:

In triangle $\triangle OTI$ (Earth center O , warship T , ionosphere I):

- Angle at T : $\frac{\pi}{2} + E$
- Angle at I : $\pi - \theta_i$ (so $\sin(\pi - \theta_i) = \sin \theta_i$)

By the **law of sines**:

$$\frac{\sin \theta_i}{R_e} = \frac{\sin \left(\frac{\pi}{2} + E \right)}{R_e + h_i} = \frac{\cos E}{R_e + h_i}$$

$$\implies \sin \theta_i(E) = \frac{R_e}{R_e + h_i} \cos E$$

The angle at earth center is:

$$\alpha(E) = \frac{\pi}{2} - E - \theta_i(E)$$

Range vs. Elevation E

$$d(E) = 2R_e \left(\frac{\pi}{2} - E - \arcsin \left(\frac{R_e \cos E}{R_e + h_i} \right) \right)$$

Physical Interpretation

As the radar antenna tilts upward ($E \uparrow$):

- θ_i decreases (steeper incidence).
- Ground range $d(E)$ decreases.
- $\cos \theta_i$ increases, requiring higher plasma density to reflect.

Derivation of the Radar "Skip/Blind Zone"

Suppose the warship operates at fixed frequency $f > f_p$ (e.g. $f = 15$ MHz).

Total internal reflection requires $\cos \theta_i \leq \frac{f_p}{f}$, which imposes a **minimum incidence angle** $\theta_{i,\min}$:

$$\cos \theta_{i,\min} = \frac{f_p}{f} \implies \sin \theta_{i,\min} = \sqrt{1 - \left(\frac{f_p}{f}\right)^2}$$

Rays launched at steeper angles ($\theta_i < \theta_{i,\min}$) escape to space. The maximum elevation angle that still reflects is:

$$\cos E_{\max} = \frac{R_e + h_i}{R_e} \sin \theta_{i,\min} = \frac{R_e + h_i}{R_e} \sqrt{1 - \left(\frac{f_p}{f}\right)^2}$$

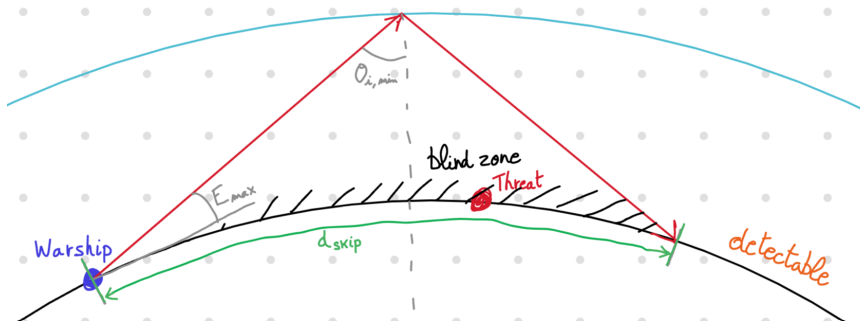
Skip/Blind Distance (d_{skip})

The minimum distance at which skywave radar returns to Earth is the **Skip Distance**:

$$d_{\text{skip}} = 2R_e \left(\frac{\pi}{2} - E_{\max} - \theta_{i,\min} \right)$$

Example of Skip Distance

Skip distance: $d_{\text{skip}} = 2R_e \left(\frac{\pi}{2} - E_{\text{max}} - \theta_{i,\text{min}} \right)$



Example, daytime ($f = 15$ MHz, $f_p = 4.02$ MHz, $h_i = 120$ km):

- $\theta_{i,\text{min}} = \arccos(4.02/15) = 74.46^\circ = 1.300$ rad
- $\cos E_{\text{max}} = \frac{6495}{6375} \sqrt{1 - (4.02/15)^2} = 1.0188 \times 0.9635 = 0.9816 \implies E_{\text{max}} = 11.0^\circ = 0.192$ rad
- $d_{\text{skip}} = 2 \times 6375 \times \left(\frac{\pi}{2} - 0.192 - 1.300 \right) = 2 \times 6375 \times 0.0788 \approx 1005$ km

Line-of-Sight vs. Tropospheric Ray Bending in the Blind Zone

Fixing the Problem

Can we detect a threat located inside the **Blind Zone** ($d < d_{\text{skip}} \approx 1000$ km) using direct Line-of-Sight (**LOS**) or **Tropospheric Ray Bending** ($k_e \approx 4/3$)?

1. **Pure Geometric Horizon** ($k_e = 1$): No atmospheric refraction ($R_{\text{eff}} = R_e = 6375$ km):

$$d_{\text{geom}} \approx \sqrt{2R_e h_{\text{mast}}} + \sqrt{2R_e h_{\text{target}}}$$

With $\sqrt{2R_e} \approx 3.571$ km/ $\sqrt{\text{m}}$:

- Warship mast: $h_{\text{mast}} = 25$ m
- Threat: $h_{\text{target}} = 10$ m

$$\begin{aligned} d_{\text{geom}} &= 3.571 \times (\sqrt{25} + \sqrt{10}) \\ &= 3.571 \times (5 + 3.162) = 29.15 \text{ km} \end{aligned}$$

2. **With Tropospheric Bending** ($k_e = 4/3$): Downward ray curvature ($R_{\text{eff}} = 8500$ km):

$$d_{\text{tropo}} \approx \sqrt{2k_e R_e} \left(\sqrt{h_{\text{mast}}} + \sqrt{h_{\text{target}}} \right)$$

With $\sqrt{2k_e R_e} \approx 4.123$ km/ $\sqrt{\text{m}}$:

$$\begin{aligned} d_{\text{tropo}} &= 4.123 \times (\sqrt{25} + \sqrt{10}) \\ &= 4.123 \times 8.162 = 33.65 \text{ km} \end{aligned}$$

Tropospheric Effect Advantage

Here ray bending extends the threat horizon by only:
 $\Delta d = 33.65 - 29.15 = +4.5$ km

Target Altitude Impact & Radar Coverage Analysis

What if the incoming threat is a high-altitude aircraft ($h_{\text{target}} = 10\,000\text{ m} = 10\text{ km}$)?

$$d_{\text{geom}} = 3.571 \times \left(\sqrt{25} + \sqrt{10\,000} \right) = 3.571 \times (5 + 100) = 375.0\text{ km}$$

$$d_{\text{tropo}} = 4.123 \times \left(\sqrt{25} + \sqrt{10\,000} \right) = 4.123 \times (5 + 100) = 432.9\text{ km}$$

Target Threat	Geometric Horizon	Tropospheric Bending		Skywave Min Distance
Sea-level ($h = 10\text{ m}$)	29.2 km	33.7 km		1005 km
Low-alt ($h = 500\text{ m}$)	97.7 km	112.8 km		1005 km
High-alt ($h = 10\text{ km}$)	375.0 km	432.9 km		1005 km

Conclusion

- Neither pure LOS nor standard tropospheric ray-bending can bridge the entire blind gap of Skywave radars.
- Need to deploy other technologies to cover this skip zone.

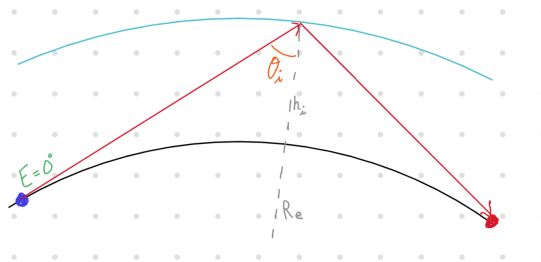
Synthesis: OTH Radar

What the radar operators need:

Property	Mathematical Formula	Significance for Naval OTH Radar
Plasma Frequency	$f_p = \frac{1}{2\pi} \sqrt{\frac{q_e^2}{m_e \epsilon_0}} \sqrt{n_e} \approx 8.98 \sqrt{n_e}$	Sets intrinsic plasma frequency.
Critical Angle	$\sin \theta_c = \sqrt{\epsilon_r} = \sqrt{1 - \frac{f_p^2}{f^2}}$	Angle threshold for total reflection ($\theta_i \geq \theta_c$).
Max Usable Frequency	$f_{\max} = \frac{f_{p,\max}}{\cos \theta_i} = f_{p,\max} \sec \theta_i$	Indicates maximum radar frequency (≤ 21 MHz).
Max Single-Hop Range	$d_{\max} = 2R_e \left(\frac{\pi}{2} - \arcsin \left(\frac{R_e}{R_e + h_i} \right) \right)$	Extends detection from 34 km to 2250 – 4650 km.
Skip/Blind Distance	$d_{\text{skip}} = 2R_e \left(\frac{\pi}{2} - E_{\max} - \theta_{i,\min} \right)$	Defines blind zone (~ 1000 km) where rays escape.
Tropospheric Horizon	$d_{\text{tropo}} = \sqrt{2k_e R_e} (\sqrt{h_T} + \sqrt{h_R})$	Bounded to 34 km for sea-level threats ($k_e = 4/3$).

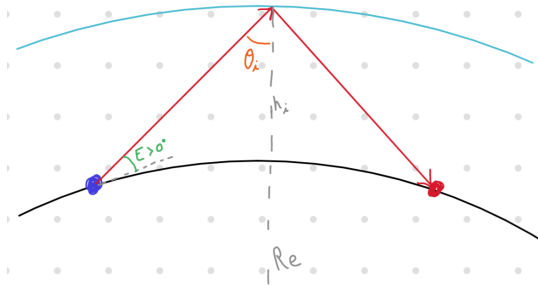
Appendix: Frequency Dependency of the Elevation

We can see that:



Grazing elevation $E = 0^\circ \implies$ bigger θ_i
 $\implies$ We can use higher frequencies and still not go over the critical angle ($\theta_i > \theta_c$).

Reach longer distances.



Higher elevation $E \implies$ smaller θ_i
 $\implies$ We have to use lower frequencies and to not go over the critical angle ($\theta_i > \theta_c$).

Reach closer distances.